Calcurion Physics Worksheets

A-Level Physics

Topic: Particles & Radiation

MARKSCHEME

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1. Gravity, electromagnetic, strong, weak **A2** [2]
2. Particle of quarks; proton, neutron (or pion, kaon) **A2** [2]
3. Fundamental particle not affected by strong force; electron, neutrino **A2** [2]
4. $p = uud$; $n = udd$ **A2** [2]
5. Same mass, opposite charge; e.g. positron **A2** [2]
6. $\gamma \rightarrow e^- + e^+$ **A2** [2]
7. $E_{\min} = 2m_e c^2 = 1.64 \times 10^{-13}$ J (1.02 MeV) **M1A1** [3]
8. $e^- + e^+ \rightarrow 2\gamma$ **A2** [2]
9. Charge / mass **A1** [1]
10. 9.58×10^7 C kg^{-1} **M1A1** [2]
11. α ($^4_2$He), β (electron/positron), γ (photon) **A3** [3]
12. $I \propto 1/r^2$ **A1** [1]
13. 2 units **A2** [2]
14. Energy gained by electron accelerated through 1 V **A1** [1]
15. 8.0×10^{-13} J **M1A1** [2]
16. $\bar{u}d$ **A2** [2]
17. Neutron heavier; energy released; proton lighter than neutron **A3** [3]
18. Energy, momentum, charge, baryon number, lepton number **A3** [3]
19. $d \rightarrow u + e^- + \bar{\nu}_e$ **A3** [3]
20. KE total $= 2.0 - 1.02 = 0.98$ MeV **M1A1** [3]
21. Changes quark flavour; responsible for β decay **A3** [3]
22. $p = \sqrt{2mE}$; $\lambda = h/p = 1.23 \times 10^{-10}$ m **M1A1** [4]
23. $hf = \phi + KE_{\max}$ **A3** [3]
24. $hf = 3.11$ eV; $KE_{\max} = 1.11$ eV **M1A1** [4]
25. Energy/momentum continuum; missing energy/momentum **A3** [3]